WANG ZHENGGANG

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EDUCATION

B.Eng. in Artificial Intelligence • Southwestern University of Finance and Economics *Chengdu, China | Sep 2023 – Jun 2027 (expected)*

- GPA: 86/100 | Rank: 35 / 196 in Department
- Relevant Coursework: Machine Learning, Deep Learning, Digital Image Processing, Machine Vision, Natural Language Processing, Data Mining, Data Structures, Graph Theory & Applications, Probability Theory, Python Programming
- Scholarships & Awards: **National Encouragement Scholarship (2025)** — China's national merit scholarship awarded to the top 3% of undergraduate students; **School-Level Scholarship (2023, 2024)**;
- CET-4 (passed-550)

PUBLICATIONS

[ICML 2026] **Wang, Z.**, Wang, W. (corresp.), Liang, H., & Jiang, T. Butterworth as Attention: Butterworth as Attention: Anisotropic Spectral Gating for Pansharpening. *International Conference on Machine Learning (ICML 2026)*. [CCF-A]

- ❖ Observation: naively transplanting spatial operators (CNN, ViT) into the frequency domain is structurally inappropriate. Each spectral coordinate encodes a specific frequency component, not a spatial location, and spectral energy follows a highly non-uniform radial distribution.
- ❖ Theoretical contribution: derived a closed-form equivalence between the classical Butterworth filter and self-attention (log cutoff frequency as Query, filter order as Key, frequency coordinates as dynamic positional encoding), replacing $O(N^2)$ Softmax attention with an $O(N \log N)$ Sigmoid-gated frequency mask. Recognizing that the standard isotropic Butterworth's circular symmetry conflicts with pansharpening's need for directional high-frequency preservation, we extended this to a learnable anisotropic filter with a dynamically oriented elliptical passband for direction-aware spectral modulation.
- ❖ Achieves SOTA performance across multiple pansharpening benchmarks with a lightweight model footprint. The core module further generalizes to CIFAR-100 image classification, outperforming spatial-domain backbones (ResNet, ConvNeXt) and the frequency-domain competitor GFNet, validating the paradigm's broader applicability.

RESEARCH EXPERIENCE

Research Assistant • Nice Lab, SWUFE *Chengdu, China | 2025 – Present*

- ❖ Conducted research under the supervision of Researcher Wang Wu.
- ❖ Key contributions:

Independently proposed the Anisotropic Butterworth Gating mechanism, serving as the core operator of ABFNet;
Designed experiments and performed ablation studies;
Authored the full manuscript and managed the complete submission pipeline for ICML 2026.

SELECTED PROJECTS

ABFNet: Anisotropic Butterworth Fusion Network (ICML 2026) • Research Implementation *| 2024 – 2026*

- ❖ Task: pansharpening — fusing high-resolution panchromatic (PAN) images with low-resolution multispectral (LMS) images to produce high-resolution multispectral output, a core problem in remote sensing with applications in environmental monitoring and urban planning.
- ❖ Architecture: a dual-branch network in which both branches operate natively in the frequency domain via FFT. The Global Branch applies a single learnable anisotropic Butterworth filter over the full feature map to capture holistic directional context; the Local Branch uses Butterworth Token Interaction (BTI) to first aggregate patch-level

semantics globally via frequency-domain filtering, then apply per-patch anisotropic filters for locally adaptive texture enhancement.

- Core operator: a learnable anisotropic Butterworth filter whose elliptical passband dynamically adapts its orientation and cutoff to the dominant texture direction — functioning as an $O(N \log N)$ frequency-domain alternative to $O(N^2)$ self-attention, with the filter implemented entirely via element-wise operations on the FFT spectrum.

Sam College — AI Agent-Powered Intelligent Education Platform | *2025 – 2026*

- A full-stack AI tutoring platform built around a ReAct Agent that classifies tasks and dynamically dispatches to 10+ domain-specific Skills and 16+ MCP tools (RAG retrieval, web search, code execution). The knowledge layer combines hybrid RAG (dense + BM25 + cross-encoder re-ranking) with structured metadata for hallucination-resistant responses; a LightGBM cold-start model predicts each new user's learning profile at onboarding, continuously refined via a three-tier memory system. Deployed with React / FastAPI, RabbitMQ async scheduling, and Docker Compose with multi-user agent isolation.

SKILLS

Programming: Python , C++, Bash/Shell

Frameworks: PyTorch, Scikit-learn, Pandas, NumPy

Tools: Git, Linux, LaTeX, Jupyter, SQL

Research: Literature review, experimental design, ablation study design, academic writing

Familiar with: VMamba, Transformer variants (ViT, Swin), GAN-based image synthesis

Languages: Mandarin (native); English